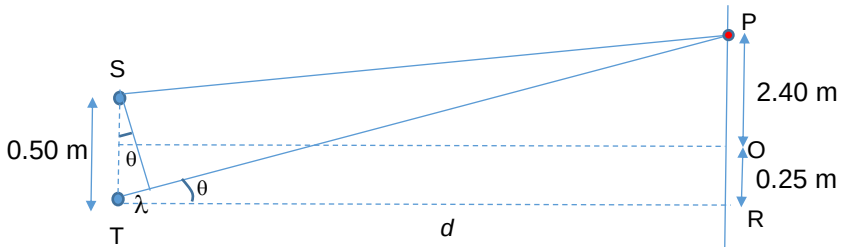


7	(a)	- The <u>direction of the oscillations</u> or vibrations is the key difference between longitudinal and transverse waves. - For longitudinal waves, the vibrations are <u>parallel</u> to the direction of wave (or energy) travel. - For transverse waves, the vibrations are <u>perpendicular</u> to the direction of wave (or energy) travel. (If student gave first point only, one mark is awarded. If student gave second and third points, full marks are awarded) Transverse wave can be polarised but longitudinal wave cannot. [B1]	B1 B1
	(b)(i)	$I = \frac{P}{4\pi D^2}$ Since intensity is proportional to the power of the sound source for fixed distance, the intensity is also halved when the power is halved Intensity is directly proportional to the square of the amplitude. $\Rightarrow I = kA^2$ $\Rightarrow \frac{I_2}{I_1} = \left(\frac{A_2}{A_1}\right)^2$ $\Rightarrow \frac{1}{2} = \left(\frac{A_{new}}{A}\right)^2$ $\Rightarrow A_{new} = \frac{A}{\sqrt{2}} \quad or \quad 0.71A$	C1 A1
	(b)(ii)	Intensity from a point source follows the inverse square law. The sound intensity $I = \frac{P}{4\pi D^2}$ For the intensity to be the same, $\frac{\frac{1}{2}P}{4\pi D'^2} = \frac{P}{4\pi D^2}$ $\Rightarrow D' = \frac{1}{\sqrt{2}}D = 0.71D$ To make the perceived intensity return back to its original intensity, one must move closer to the sound source.	C1 A1

	(c)(i) The two waves are coherent because they have a <u>constant phase difference</u> (or constant phase relation) of π radian or 180°. or From Fig. 7.2, the periods of both waves are same at ALL times. (Value of phase difference is not necessary) (Reject the following notions - having same frequency, period, speed etc. Two independent sources may have the same frequency but are not coherent because their phase relation may not be the same at all times.)	B1
	(c)(ii) The two waves arrive at P <u>in antiphase (180°) to each other</u> (crest of one meets the trough of the other wave) and <u>so destructive interference takes place</u>. However, it is only a partial cancellation as <u>their amplitudes are not the same (they do not cancel completely)</u>. Hence, sound intensity at P is not zero but is a minimum.	B1 B1
	(c)(iii) The resultant intensity is I. No explanation is required here. After superposition, the resultant amplitude is still that of S alone. Thus, the intensity is the same as S alone.	B1
	(c)(iv) From Fig. 8.2, period $T = 0.800$ ms, $c = f\lambda$ $330 = \left(\frac{1}{0.800 \times 10^{-3}}\right)\lambda$ $\lambda = 0.264 \text{ m}$	M1 A1
	(d)(i) P and O are successive minima and so the path difference must be one wavelength, $\lambda = 0.264$ m	B1
	(d)(ii) 1.  Since P is the first minimum after O, the path difference $(TP - SP) = \lambda$ Thus, $\sin\theta = \frac{\lambda}{a} = \frac{0.264}{0.50} = 0.528$ $\Rightarrow \theta = 31.9^\circ$	C1 A1
	(d)(ii) 2. Looking at triangle TPR, we have $PR = PO + OR = (2.40 + 0.25) \text{ m} = 2.65$	C1

9	(a)(i)	To remove air from the chamber so that alpha particles	P1
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